Algebraic Curves
Chapter 7.5. Nonsingular Models of Curves

Proposition.
Let $\mathbb{C}$ be a projective curve. Then there exists a nonsingular projective curve $\mathbb{X}$ and birational morphism $f\colon \mathbb{X} \to \mathbb{C}$. Moreover if $f'\colon \mathbb{X}' \to \mathbb{C}$ is another such birational morphism, then there is a unique isomorphism $g\colon \mathbb{X} \to \mathbb{X}'$ s.t. $f' \circ g = f$.

Proof.

Uniqueness follows from Chapter 7.1. Corollary 2. For existence recall that all varieties of dimension two are birationally equivalent to a plane curve. Then by quadratic transformations w.l.o.g. let the only singularities of this plane curve be ordinary. The end result of Chapter 7.3. therefore states that this curve is birationally equivalent to a nonsingular curve $\mathbb{X}$. That the composition of all these birational maps yields a morphism follows from Chapter 7.3. Corollary 1. Now if $\mathbb{C}$ is a plane curve, then f mapping $\mathbb{X}$ onto $\mathbb{C}$ is a result of the construction of $\mathbb{X}$ from $\mathbb{C}$. Namely let $\mathbf{p} \in \mathbb{C}$. Then we may construct $\mathbb{C}'$ with only ordinary singularities s.t. the rational map $\mathbb{C}' \to \mathbb{C}$ is defined at some $\mathbf{p}'$ and $\mathbf{p}' \mapsto \mathbf{p}$, with $\mathbb{X}$ surjecting onto $\mathbb{C}'$. Thus let $\mathbb{C} \subseteq \mathbb{P}^n$ and $\mathbf{p} \in \mathbb{C}$. Choose our morphism $g\colon \mathbb{C} \to \mathbb{C}_1$ s.t. $\mathbf{p} = g^{-1}\, g[\mathbf{p}]$ (using Question 6.43). Thus if $g\, f[\mathbf{x}] = \mathbf{p}$, then $f[\mathbf{x}] = \mathbf{p}$.

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As a corollary, there is a natural correspondence between nonsingular projective curves $\mathbb{X}$ and algebraic function fields in one variable $\mathbb{K}$ over $\Bbbk$ in $\mathbb{K} = \Bbbk(\mathbb{X})$. If $\mathbb{X}$ and $\mathbb{X}'$ are two such curves, then dominant morphisms between $\mathbb{X}'$ and $\mathbb{X}$ are homomorphisms between $\Bbbk(\mathbb{X})$ and $\Bbbk(\mathbb{X}')$.

Let $\mathbb{C}$ be a projective curve and $f\colon \mathbb{X} \to \mathbb{C}$ as in the proposition. $\mathbb{X}$ is known as the 'nonsingular model' of $\mathbb{C}$, or of $\mathbb{K} = \Bbbk(\mathbb{C})$. $\Bbbk(\mathbb{X})$ is related to $\mathbb{K}$ using f_*. The points $\mathbf{q} \in \mathbb{X}$ are in correspondence to the d.v.r. $\mathscr{O}_{\mathbf{q}}[\mathbb{X}]$ of $\mathbb{K}$ using Chapter 7.1. Corollary 4. Hence $f[\mathbf{q}] = \mathbf{p}$ *iff* $\mathscr{O}_{\mathbf{q}}[\mathbb{X}]$ dominates $\mathscr{O}_{\mathbf{p}}[\mathbb{C}]$. Points of $\mathbb{X}$ are therefore called 'places' of $\mathbb{C}$ or $\mathbb{K}$, and a place $\mathbf{q}$ is 'centered' at $\mathbf{p}$ if $f[\mathbf{q}] = \mathbf{p}$.

Proposition.
Let $\mathbb{C}$ be a projective plane curve and $\mathbf{p} \in \mathbb{C}$. Then there is an affine neighbourhood $\mathbb{U}$ about $\mathbf{p}$ s.t.,

1. $\mathrm{f}^{-1}[\mathbb{U}] =: \mathbb{U}'$ is an affine open subvariety of $\mathbb{X}$
2. $\Bbbk[\mathbb{U}']$ is module-finite over $\Bbbk[\mathbb{U}]$
3. There exists some $t \in \Bbbk[\mathbb{U}] \smallsetminus 0$ s.t. $t\Bbbk[\mathbb{U}'] \subseteq \Bbbk[\mathbb{U}]$.
4. The vector space $\Bbbk[\mathbb{U}']/\Bbbk[\mathbb{U}]$ is finite-dimensional over $\Bbbk$.

In particular our neighbourhood may exclude any finite set S of points in $\mathbb{C}$ if $\mathbf{p} \notin S$.

Proof.

We choose successive quadratic transformations $\mathbb{C} = \mathbb{C}_1, \ldots, \mathbb{C}_n$ so that $\mathbb{C}_n$ has only ordinary singularities, and likewise open affine sets $\mathbb{W}_i \subseteq \mathbb{C}_i$ s.t. $\mathbf{p} \in \mathbb{W}_1$ and $S \cap \mathbb{W}_1 = \varnothing$, the birational map $\mathbb{C}_{i+1} \to \mathbb{C}_i$ is represented by the morphism $\mathrm{f}_i \colon \mathbb{W}_{i+1} \to \mathbb{W}_i$, and the maps f_i all satisfy the antecedents of Chapter 7.2, Step 3. This is possible by Chapter 7.2 and Question 7.13. Lastly let $\mathrm{f}_n \colon \mathbb{X} \to \mathbb{C}_n$ be the nonsingular model of $\mathbb{C}_n$ and $\mathbb{W}_{n+1} = \mathrm{f}^{-1}[\mathbb{W}_n]$. This is possible by Chapter 7.3. Step 7. Define $\mathbb{U} := \mathbb{W}_1$ and $\mathbb{U}' := \mathbb{W}_{n+1}$. By Question 1.45 we have that $\Bbbk[\mathbb{U}']$ is module-finite over $\Bbbk[\mathbb{U}]$. Write,

$$\Bbbk[\mathbb{U}'] = \sum_{i=1}^{m} \Bbbk[\mathbb{U}] z_i$$

Then as $\Bbbk[\mathbb{U}]$ and $\Bbbk[\mathbb{U}']$ have the same quotient fields we find that there is $t \in \Bbbk[\mathbb{U}]$ with $tz_i \in \Bbbk[\mathbb{U}]$ for $t \neq 0$. This is our t. Lastly define,

$$\varphi \colon \Bbbk[\mathbb{U}']/\Bbbk[\mathbb{U}] \to \Bbbk[\mathbb{U}]/\langle t \rangle \qquad \bar{z} \mapsto t\bar{z}$$

Observe that φ is an injective $\Bbbk$-linear map, so to prove finite-dimensionality it suffices to prove that $\Bbbk[\mathbb{U}]/\langle t \rangle$ is finite-dimensional. This is as t may only have finitely many zeroes in $\mathbb{U}$ from Chapter 1.7. Corollary 4.

■

For the following let $\mathrm{f} \colon \mathbb{X} \to \mathbb{C}$ for $\mathbf{q} \in \mathbb{X}$, $\mathrm{f}[\mathbf{q}] = \mathbf{p} \in \mathbb{C}$. For any plane curve $\mathbb{G}$ we have $\mathbb{G}_* \in \mathscr{O}_{\mathbf{p}}[\mathbb{P}^2]$ by dehomogenizing in Z. Denote $\eth$ to be the image of $\mathbb{G}_*$ in $\mathscr{O}_{\mathbf{p}}[\mathbb{C}] \subseteq \Bbbk(\mathbb{C}) = \Bbbk(\mathbb{X})$. Define $\mathrm{ord}_{\mathbf{q}}[\mathbb{G}] = \mathrm{ord}_{\mathbf{q}}[\eth]$.

Proposition.
Let $\mathbb{C}$ be an irreducible projective plane curve, $\mathbf{p} \in \mathbb{C}$, and $\mathrm{f} \colon \mathbb{X} \to \mathbb{C}$. Let $\mathbb{G}$ be a possibly reducible plane curve. Then,

$$\langle \mathbb{C}, \mathbb{G} \rangle[\mathbf{p}] = \sum_{\mathbf{q} \in \mathrm{f}^{-1}[\mathbf{p}]} \mathrm{ord}_{\mathbf{q}}[\mathbb{G}]$$

Proof.

By the previous proposition we have a neighbourhood $\mathbb{U}$ about $\mathbf{p}$, and namely we may shrink it so that $\eth$ is unit in $\mathscr{O}_{\mathbf{p}'}[\mathbb{C}]$ for all $\mathbf{p}' \in \mathbb{U} \smallsetminus \{\mathbf{p}\}$. We first have,

$$\langle \mathbb{C}, \mathbb{G} \rangle[\mathbf{p}] = \dim_{\Bbbk}[\mathscr{O}_{\mathbf{p}}[\mathbb{P}^2]/\langle \mathrm{F}_*, \mathrm{G}_* \rangle] = \dim_{\Bbbk}[\mathscr{O}_{\mathbf{p}}[\mathbb{C}]/\langle \eth \rangle] = \dim_{\Bbbk}[\Bbbk[\mathbb{U}]/\langle \eth \rangle]$$

Define,

$$\mathrm{T}\colon \Bbbk[\mathbb{U}] \to \Bbbk[\mathbb{U}'], \qquad \mathrm{z} \mapsto \mathrm{g}\,\mathrm{z}$$

Now $\Bbbk[\mathbb{U}']/\Bbbk[\mathbb{U}]$ is finite-dimensional so we have $\dim \Bbbk[\mathbb{U}]/\,\mathrm{T}[\Bbbk[\mathbb{U}]] = \dim \Bbbk[\mathbb{U}']/\,\mathrm{T}[\Bbbk[\mathbb{U}']]$. Hence $\dim_{\Bbbk} \Bbbk[\mathbb{U}]/\langle \Bbbk \rangle = \dim \Bbbk[\mathbb{U}']/\langle \eth \rangle$. Therefore,

$$\dim \Bbbk[\mathbb{U}']/\langle \eth \rangle = \sum_{\mathbf{q} \in \mathrm{f}^{-1}[\mathbf{p}]} \dim \mathscr{O}_{\mathbf{q}}[\mathbb{X}]/\langle \eth \rangle = \sum_{\mathbf{q}} \mathrm{ord}_{\mathbf{q}}\, \mathrm{g}$$

As desired.

■

Proposition.
Let $\mathbf{p}$ be an ordinary singularity of $\mathbb{C}$ of multiplicity r. Write $\mathrm{f}^{-1}[\mathbf{p}] = \{\mathbf{p}_1, \ldots, \mathbf{p}_r\}$. If $\mathrm{z} \in \Bbbk(\mathbb{C})$ and $\mathrm{ord}_{\mathbf{p}_i} \geq r - 1$, then $\mathrm{z} \in \mathscr{O}_{\mathbf{p}}[\mathbb{C}]$.

Proof.

Let $\mathbb{U}$ be the neighbourhood about $\mathbf{p}$ as sufficiently small so that $\mathrm{z} \in \mathscr{O}_{\mathbf{p}'}[\mathbb{C}]$ for all $\mathbf{p}' \in \mathbb{U} \smallsetminus \{\mathbf{p}\}$, and s.t. $\mathrm{f}\colon \mathbb{U}' := \mathrm{f}^{-1}[\mathbb{U}] \to \mathbb{U}$ satisfies the antecedents of Chapter 7.2. Step 3. From Step 2 of that step therefore $\mathrm{ord}_{\mathbf{p}_i}[x] = 1$ so that $\mathrm{z}\, x^{1-r} \in \Bbbk[\mathbb{U}']$. However $x^{r-1}\Bbbk[\mathbb{U}'] \subseteq \Bbbk[\mathbb{U}]$, so $z = x^{r-1}(\mathrm{z}\, x^{1-r}) \in \Bbbk[\mathbb{U}] \subseteq \mathscr{O}_{\mathbf{p}}[\mathbb{U}]$.

■

Proposition.
Let $\mathbb{F}$ be an irreducible projective plane curve and $\mathbf{p}$ an ordinary singularity of multiplicity r on it. Let $\mathbf{p}_1, \ldots, \mathbf{p}_r$ be places centered at $\mathbf{p}$. Let $\mathbb{G}$, $\mathbb{H}$ be plane curves. Then Noether's conditions are satisfied at $\mathbf{p}$ if $\mathrm{ord}_{\mathbf{p}_i}[\mathbb{H}] \geq \mathrm{ord}_{\mathbf{p}_i}[\mathbb{G}] + r - 1$ for all i.

Proof.

Observe that $\mathbb{H}_* \in \langle \mathbb{F}_*, \mathbb{H}_* \rangle \subseteq \mathscr{O}_{\mathbf{p}}[\mathbb{P}^2]$ is equivalent to $\bar{\mathbb{H}}_* \in \langle \bar{\mathbb{G}}_* \rangle \subseteq \mathscr{O}_{\mathbf{p}}[\mathbb{F}]$. Namely this is also equivalent to $\mathrm{z} = \bar{\mathbb{H}}_*/\bar{\mathbb{G}}_* \in \mathscr{O}_{\mathbf{p}}[\mathbb{F}]$. The previous proposition thus gives the result.

■

Question 7.17

Part A

Proposition.
For every irreducible curve $\mathbb{C}$ there is a nonsingular curve $\mathbb{X}$ and a birational morphism $\mathbb{X} \to \mathbb{C}$.

Proof.

Recall that for every curve $\mathbb{C}$ there is a projective curve $\mathbb{C}' \subseteq \mathbb{P}^2$ that is birationally equivalent to it by a map $g\colon \mathbb{C}' \to \mathbb{C}$. Then by the results of this chapter there is a nonsingular model $\mathbb{X}$ of $\mathbb{C}'$, and a birational morphism $f : \mathbb{X} \to \mathbb{C}'$. Therefore $g \circ f\colon \mathbb{X} \to \mathbb{C}$ is a birational map. As $\mathbb{X}$ is nonsingular, by Chapter 7.1. Corollary 1 $g \circ f$ is a birational morphism.

By the injective correspondence of *projective* nonsingular curves and algebraic function fields we see that $\Bbbk(\mathbb{C})$ corresponds to a *projective* nonsingular curve $\mathbb{X}$. Now $\Bbbk(\mathbb{C})$ is the unique function field of $\mathbb{C}$, so that the choice of $\mathbb{X}$ is unique (up to isomorphism).

Part B

Proposition.
Let $f\colon \mathbb{X} \to \mathbb{C}$ be the aforementioned birational morphism and $\mathbb{C}^0$ the set of simple points of $\mathbb{C}$. Then f restricted to $f^{-1}[\mathbb{C}^0]$ is an isomorphism of $\mathbb{C}^0$ with $f^{-1}[\mathbb{C}^0]$.

Proof.

W.l.o.g. by isomorphism let $\mathbb{C}$ be projective. Between $\mathbb{C}$ and $\mathbb{X}$ we observe that $f\colon \mathbb{X} \to \mathbb{C}$ is a rational map from a nonsingular curve to a projective curve, so that the domain of f includes all simple points of $\mathbb{X}$ and f is a morphism. In particular by restriction $f\colon f^{-1}[\mathbb{C}^0] \to \mathbb{C}$ is a morphism. Now observe in the other direction that $f^{-1}\colon \mathbb{C} \to \mathbb{X}$ is also a rational map from a nonsingular curve to a projective curve, so its domain includes all simple points of $\mathbb{C}$, and in particular its restriction $f^{-1}\colon \mathbb{C}^0 \to \mathbb{X}$ is a morphism. In conjunction $f\colon f^{-1}[\mathbb{C}^0] \to \mathbb{C}^0$ is an isomorphism.

■

Question 7.18

Proposition.
For a place $\mathbf{p}$ of $\mathbb{C}$ and any choice t of uniformizing parameter of $\mathscr{O}_{\mathbf{p}}[\mathbb{X}]$, there is a homomorphism $\Bbbk(\mathbb{X}) \to \Bbbk((T))$.

Proof.

It suffices to observe that $\mathscr{O}_{\mathbf{p}}[\mathbb{X}]$ is a d.v.r. with maximal ideal $\langle \mathbf{x} - \mathbf{p} \rangle$, quotient field $\Bbbk(\mathbb{X})$, and containing the field $\Bbbk$. Now for the homomorphism $\mathrm{eval}_{\mathbf{x}=\mathbf{p}} \colon \mathscr{O}_{\mathbf{p}}[\mathbb{X}] \to \Bbbk$ we have by the First Isomorphism Theorem,

$$\mathscr{O}_{\mathbf{p}}[\mathbb{X}]/\langle \mathbf{x} - \mathbf{p} \rangle \cong \Bbbk$$

So that the composition $\Bbbk \hookrightarrow \mathscr{O}_{\mathbf{p}}[\mathbb{X}] \to \mathscr{O}_{\mathbf{p}}[\mathbb{X}]/\langle \mathbf{x} - \mathbf{p} \rangle$ is an isomorphism. Therefore by Question 2.30-32 each $z \in \Bbbk(\mathbb{X})$ has a power series expansion,

$$z = \sum_{i=0}^{\infty} \lambda_i T^i$$

Where this is a homomorphism $\Bbbk(\mathbb{X}) \to \Bbbk((T))$, and the uniformizing parameter t maps to T.

To recover $\mathbf{p}$ from this map simply consider the preimage of T under this map, which would give the uniformizing parameter t, which is generically of the form $(\mathbf{x} - \mathbf{p})\, u$ for some invertible $u \in \mathscr{O}_{\mathbf{p}}[\mathbb{X}]$. Therefore we may locate $\mathbf{p}$ from the uninvertible part.

■

Question 7.19

Proposition.
Let $\mathrm{f}\colon \mathbb{X} \to \mathbb{C}$ be a birational morphism for $\mathbb{C}$ a projective plane curve. Let $\mathbf{p}$ be the ordinary singularity of multiplicity r on $\mathbb{C}$ and $\mathbf{q}_1, \ldots, \mathbf{q}_r$ the places on $\mathbb{X}$ centered at $\mathbf{p}$. Let $\mathbb{G}$ be any other projective plane curve and $s \leq r$. Then $\mathrm{m}_{\mathbf{p}}[\mathbb{G}] \geq s$ *iff* $\mathrm{ord}_{\mathbf{q}_i}[\mathbb{G}] \geq s$ for all i.

Proof.

$(\Rightarrow)$

Let $\mathrm{m}_{\mathbf{p}}[\mathbb{G}] \geq s$. Recall that about $\mathbf{p}$ there is an affine neighbourhood $\mathbb{U} \subseteq \mathbb{C}$ s.t. $\mathbb{U}' := \mathrm{f}^{-1}[\mathbb{U}] \subseteq \mathbb{X}$ is affine. We restrict to this neighbourhood. Let us first note that as $\mathscr{O}_{\mathbf{q}_i}[\mathbb{X}]$ is a d.v.r. that $\mathscr{O}_{\mathbf{q}_i}[\mathbb{U}']$ is also a d.v.r. which inherits the order function being a restriction to an affine neighbourhood. From the birational morphism $\mathbb{X} \to \mathbb{C}$ we have using its restriction and its pullback the inclusion $\Bbbk[\mathbb{U}] \subseteq \Bbbk[\mathbb{U}']$. Therefore by Question 7.5,

$$\mathrm{ord}_{\mathbf{q}_i}[\mathbb{G}] = \mathrm{ord}_{\mathbf{q}_i}^{\mathbb{U}'}[\pi_{\langle \mathbb{U} \rangle}\mathbb{G}_*] \geq \mathrm{m}_{\mathbf{p}}[\mathbb{G}] \geq s$$

$(\Leftarrow)$

Let $\mathrm{ord}_{\mathbf{q}_i}[\mathbb{G}] \geq s \geq r$. Again observe that there is an affine neighbourhood $\mathbb{U} \subseteq \mathbb{C}$ s.t. $\mathbb{U}' := \mathrm{f}^{-1}[\mathbb{U}] \subseteq \mathbb{X}$ is affine. About these neighbourhoods f is an affine morphism $\mathbb{U}' \to \mathbb{U} \Leftrightarrow \Bbbk[\mathbb{U}] \to \Bbbk[\mathbb{U}']$. Therefore by Question 7.5. $\mathrm{m}_{\mathbf{p}}[\mathbb{G}] \geq s$.

■

Question 7.20

Part A

Proposition.
Let $\mathfrak{R}$ be a domain with quotient field $\mathbb{K}$. Define the 'integral closure' $\mathfrak{R}'$ of $\mathfrak{R}$ to be the set of elements of $\mathbb{K}$ integral over $\mathfrak{R}$. Then if $\mathfrak{R}$ is a d.v.r., $\mathfrak{R}' = \mathfrak{R}$.

Proof.

The inclusion $\mathfrak{R} \subseteq \mathfrak{R}'$ is immediate as each element $r \in \mathfrak{R}$ is the root of $x - r \in \mathfrak{R}[x]$. We prove the opposite direction. Let $a/b \in \mathfrak{R}'$ be arbitrary, so is the root of some monic polynomial $\sum_i c_i x^i + x^n \in \mathfrak{R}[x]$. Given that $\mathfrak{R}$ is a d.v.r. there is some uniformizing parameter $t \in \mathfrak{R}$ s.t. all coefficients of the polynomial may be written $c_i = u_i t^{k_i}$ for invertibles $u_i \in \mathfrak{R}$ and $k_i \in \mathbb{N}$, and also $a/b = ut^k$. Therefore we have the equality,

$$0 = \sum_i u_i t^{k_i} (ut^k)^i + (ut^k)^n = \sum_i (u_i u^i) t^{k_i + ik} + u^n t^{nk}$$

We move the monic term to the l.h.s. Thus by equality of order,

$$nk = \operatorname{ord}\left[u^n t^{nk}\right] = \operatorname{ord}\left[\sum_i (u_i u^i) t^{k_i + ik}\right] \geq \min_i \operatorname{ord}\left[(u_i u^i) t^{k_i + ik}\right] = \min_i \{k_i + ik\}$$

In particular let j be where the minimum is attained. Then we have,

$$nk \geq k_j + jk \Rightarrow k \geq \frac{k_j}{n - j}$$

Where $k_j \geq 0$ as each coefficient is in $\mathfrak{R}$. Therefore $k \geq 0$, and so $a/b \in \mathfrak{R}$. Thus $\mathfrak{R}' \subseteq \mathfrak{R}$ and therefore $\mathfrak{R}' = \mathfrak{R}$.

■

Part B

Proposition.
If $\mathfrak{R}'_\alpha = \mathfrak{R}_\alpha$ then $(\bigcap \mathfrak{R}_\alpha)' = (\bigcap \mathfrak{R}_\alpha)$

Proof.

The inclusion $(\bigcap \mathfrak{R}_\alpha) \subseteq (\bigcap \mathfrak{R}_\alpha)'$ is again immediate. Elements of $(\bigcap \mathfrak{R}_\alpha)'$ are elements r of $\operatorname{Frac} \bigcap \mathfrak{R}_\alpha$ that are roots of monic polynomials $\sum_i c_i x^i + x^n$ over $\bigcap \mathfrak{R}_\alpha[x]$, so $c_i \in \mathfrak{R}_\alpha$ for any α. Viewing this polynomial as an element of $\mathfrak{R}_\alpha[x]$, therefore r is in $\mathfrak{R}'_\alpha$ as well for any α. By hypothesis r is in $\mathfrak{R}_\alpha$. As this is true for arbitrary α therefore r is in $\bigcap \mathfrak{R}_\alpha$. Therefore $(\bigcap \mathfrak{R}_\alpha)' \subseteq (\bigcap \mathfrak{R}_\alpha)$ and in concluding $(\bigcap \mathfrak{R}_\alpha)' = (\bigcap \mathfrak{R}_\alpha)$.

■

Part C

Proposition.
Let $\mathrm{f}\colon \mathbb{X} \to \mathbb{C}$ be a nonsingular model. Then $\Bbbk[\mathrm{f}^{-1}[\mathbb{U}]] = \Bbbk[\mathbb{U}]'$ for all open sets $\mathbb{U} \subseteq \mathbb{C}$.

Proof.

Denote $\mathbb{U}' := \mathrm{f}^{-1}[\mathbb{U}]$. We have the result that for some nontrivial $\mathrm{t} \in \Bbbk[\mathbb{U}]$ that $\mathrm{t}\,\Bbbk[\mathbb{U}'] \subseteq \Bbbk[\mathbb{U}]$, so for all $\mathrm{z} \in \Bbbk[\mathbb{U}']$ we have $\mathrm{t}\,\mathrm{z} \in \Bbbk[\mathbb{U}]$. Therefore $\mathrm{t}\,x - \mathrm{t}\,\mathrm{z}$ is a polynomial of $\Bbbk[\mathbb{U}][x]$, and its root is z, so $\mathrm{z} \in \Bbbk[\mathbb{U}]'$. Hence $\Bbbk[\mathbb{U}'] \subseteq \Bbbk[\mathbb{U}]'$. For the opposite direction observe that f is an birational morphism $\mathbb{U}' \to \mathbb{U}$, so the pullback f^* is a monomorphism $\Bbbk[\mathbb{U}] \hookrightarrow \Bbbk[\mathbb{U}']$. Now $\mathbb{X} \supseteq \mathbb{U}'$ is a nonsingular curve so all its points a are simple, and therefore $\mathscr{O}_a[\mathbb{U}']$ are d.v.r. for all a. By Part A they are integrally closed. Now $\Bbbk[\mathbb{U}'] = \bigcap_{a \in \mathbb{U}'} \mathscr{O}_a[\mathbb{U}']$, so it follows by Part B that it too is integrally closed. Hence,

$$\Bbbk[\mathbb{U}]' \leq \Bbbk[\mathbb{U}']' = \Bbbk[\mathbb{U}']$$

In conjunction $\Bbbk[\mathbb{U}'] = \Bbbk[\mathbb{U}]'$.

■

Question 7.21

Part A

Proposition.
Let $\mathbb{X}$ be a projective nonsingular curve and $\mathbf{p}_1, \ldots, \mathbf{p}_s \in \mathbb{X}$. Then there is a projective plane curve $\mathbb{C}$ with only ordinary multiple points and a birational morphism $f\colon \mathbb{X} \to \mathbb{C}$ s.t. $f[\mathbf{p}_i]$ is simple on $\mathbb{C}$ for all i.

Proof.

Given that $\mathbb{X}$ is a curve, i.e. a variety of dimension one, there is an affine variety $\mathbb{C}' \subseteq \mathbb{A}^2$ s.t. $\Bbbk(\mathbb{C}') \cong \Bbbk(\mathbb{X})$. By quasi-projective in $\mathbb{A}^2$ we may embed this variety into the projective plane $\mathbb{P}^2$. This curve may have nonordinary singularities. In this case we employ successive quadratic transformations on the space until the all singularities are ordinary. The resultant curve is birational to $\mathbb{X}$ through a map g. We additionally employ resolution of singularities on the image $g[\mathbf{p}_i]$, giving us a curve $\mathbb{C}$ that is birational to $\mathbb{X}$ through composition yielding f. From the quadratic maps we have that singularities on $\mathbb{C}$ are ordinary, and by resolving at $g[\mathbf{p}_i]$ we guarentee that $f[\mathbf{p}_i]$ are each simple.

■

Part B

Proposition.
For any $m_1, \ldots, m_r \in \mathbb{Z}$ there is $z \in \Bbbk(\mathbb{X})$ s.t. $\operatorname{ord}_{\mathbf{p}_i}[z] = m_i$.

Proof.

Observe that $\mathbb{C}$ from Part A is a irreducible projective plane curve s.t. $\Bbbk(\mathbb{C}) \cong \Bbbk(\mathbb{X})$, and that $\mathbf{p}_1, \ldots, \mathbf{p}_r$ are all simple points on $\mathbb{C}$. We are in a position to employ Question 5.15 to define,

$$z = \ell_0^{-\sum_i m_i} \prod_i \ell_i^{m_i}$$

Where the ℓ_i are lines transversal to $\mathbb{C}$ at each $\mathbf{p}_i$. Then z is s.t. it is an element of $\Bbbk(\mathbb{C})$ and $\operatorname{ord}_{\mathbf{p}_i}[z] = m_i$ for all i. Its image under the isomorphism into $\Bbbk(\mathbb{X})$ is our function.

■

Part C

Proposition.
The curve $\mathbb{C}$ of Part A may of arbitrarily large degree.

Proof.

By repeatedly quadratically transforming about a simple point on $\mathbb{C}$ we inflate its degree to arbitrarily high levels while preseving its genus, and so thus still has zero nonordinary singularities.

Question 7.22

Part A

Proposition.
Let $\mathbf{p}$ be a node on an irreducible plane curve F. Let ℓ_1, ℓ_2 be its tangents at $\mathbf{p}$. Supposing $\mathbf{p}$ is a simple node, then $\langle \mathrm{F}, \mathrm{H} \rangle[\mathbf{p}] = 6$.

Proof.

Given that we have a node at $\mathbf{p}$, we have the result that $\mathrm{H} \neq 0$. W.l.o.g. using a projective automorphism let $\mathbf{p} = [0 : 0 : 1]$ and $\ell_1 = x, \ell_2 = y$ so that,

$$\mathrm{f} := \mathrm{F}_* = xy + dx^3 + ex^2y + fxy^2 + gy^3 + \cdots$$

With it being a simple node saying that $d, g \neq 0$. Let $\varphi\colon \mathbb{X} \to \mathbb{C}$ be the resolution of our plane curve. Then the singularity at $\mathbf{p}$ corresponds to the two branches about $\mathbf{q}_1$ and $\mathbf{q}_2$ on $\mathbb{X}$ that are d.v.r.. Therein, using Question 5.23,

$$\langle \mathrm{F}, \mathrm{H} \rangle[\mathbf{p}] = \langle \mathrm{F}, \mathrm{G} \rangle = \mathrm{ord}_{\mathbf{q}_1}[\mathrm{G}] + \mathrm{ord}_{\mathbf{q}_2}[\mathrm{G}]$$

Where $\mathrm{G} = \partial_y^2 \mathrm{f}\, \partial_{xx} \mathrm{f} + \partial_x^2 \partial_{yy} \mathrm{f} - 2\partial_x \mathrm{f}\, \partial_y \mathrm{f}\, \partial_{xy} \mathrm{f}$. On the branch $\mathscr{O}_{\mathbf{q}_1}[\mathbb{X}]$ we have a uniformizing parameter t_1 so that $x = ut_1^n, y = vt_2^m$. Pulling back,

$$\mathrm{f} = ut_1^n + vt_2^m + \mathsf{h.o.t.}$$

So that for the vanishing the degrees of the lowest order terms must agree, at which we see $n = m = 1$. The same can be said of the uniformizing parameter t_2 of $\mathscr{O}_{\mathbf{q}_2}[\mathbb{X}]$. Now explicitly,

$$\mathrm{G} = -2xy - 8ex^2y - 8fxy^2 + \cdots$$

So that,

$$\pi_{\langle \mathrm{F} \rangle} \mathrm{G} = 2dx^3 + 2gy^3 - 7x^2y - 7xy^2 + \cdots$$

And thus its order in $\mathbf{q}_1$ and $\mathbf{q}_2$ are both three. Hence,

$$\mathrm{ord}_{\mathbf{q}_1}[\mathrm{G}] + \mathrm{ord}_{\mathbf{q}_2}[\mathrm{G}] = 3 + 3 = 6$$

As desired.

■

Part B

Proposition.
If F has an ordinary cusp at $\mathbf{p}$, then $\langle \mathrm{F}, \mathrm{H} \rangle[\mathbf{p}] = 8$.

Proof.

Recall from Question 7.6 that if $\mathbf{p}$ is an ordinary cusp of F, then $\varphi^{-1}[\mathbf{p}] = \{\mathbf{p}_1\}$ where $\mathbf{p}_1$ is simple on $\mathbb{X}$. Namely recall that this configuration arises when $\mathbf{p}$ is a double point with only one tangent, which w.l.o.g. using a projective automorphism again we may have $\mathbf{p} = [0:0:1]$ and,

$$\mathrm{F} = y^2 + x^3 + \text{h.o.t.}$$

Our normalization produces one branch. A computation yields,

$$\mathrm{G} = 18x^4 + 24xy^2 + \text{h.o.t.}$$

We pull this curve back to our normalization. On the branch of our point there is a uniformizing parameter t s.t. $x = ut^n$ and $y = vt^m$. Under this,

$$\mathrm{F} = v^2t^{2m} + u^3t^{3n} + \text{h.o.t.}$$

For this to vanish the lowest degree terms' order must agree. Hence $2m = 3n$, so $m = 3, n = 2$, and $x = ut^2, y = vt^3$. Thereby,

$$\mathrm{ord}_{\mathbf{p}_1}[\mathrm{G}] = \mathrm{ord}_{\mathbf{p}_1}[18u^4t^8 + 24uv^2t^8 + \text{h.o.t.}] = 8$$

And thus,

$$\langle \mathrm{F}, \mathrm{H}\rangle[\mathbf{p}] = \langle \mathrm{F}, \mathrm{G}\rangle[\mathbf{p}] = \mathrm{ord}_{\mathbf{p}_1}[\mathrm{G}] = 8$$

■

Part C

Proposition.
Every cubic has one, three, or nine flexes

Proof.

Note that the degree of the Hessian of a cubic is three. From Bezout's Theorem the sum of intersection numbers with a cubic and its Hessian must be nine. Recall from Question 5.13 that an irreducible cubic is either nonsingular or has at most one double point, that being a node or a cusp. If it is irreducible then by Question 5.23 it has nine flex. If it has a node then Part A showed that its intersection number with its Hessian is six. As its other points cannot be multiple points they must be flexes. In particular there exists three such points. If it has a cusp then Part B showed that its intersection number with its Hessian there is eight. As its other points cannot be multiple points they must be flexes. In particular there exists one such point.

■

Part D

Proposition.
If a curve F has degree n and i flexes, all ordinary, δ simple nodes, and k cusps with no other singularities, then,

$$i + 6\delta + 8k = 3n(n-2)$$

Proof.

The degree of the Hessian is $3(n-2)$. By Bezout's Theorem and the previous results (which in effect weight each singularity),

$$3n(n-2) = \sum_{\mathbf{p}} \langle \mathrm{F}, \mathrm{H} \rangle [\mathbf{p}] = i + 6\delta + 8k$$

■